

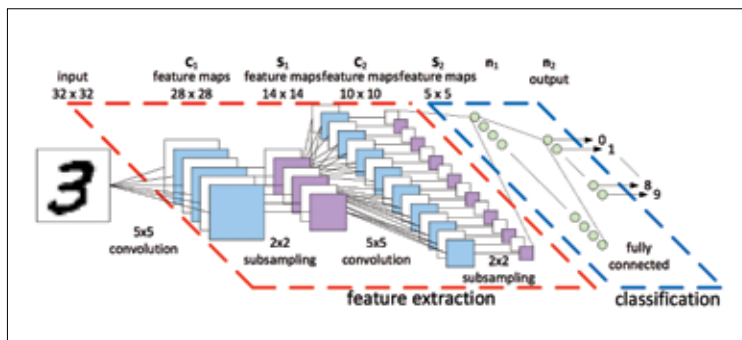
Training data:- Analysts need a lot of training data, to begin with, if they need the output to hold some consistency with the input. The conventional neural network that works with numbers requires megabytes worth of data, but CNN requirements can jump up to gigabytes and even terabytes (1024 gigabytes). Analysts can always use smaller sets but run the demerit of losing dimensional consistency.

Data Collection:- Data collection no doubt becomes tougher for CNNs if the program requires good results and verifications.

Data Verification:- Obtaining results is just one part of the puzzle but verifying them against the input to check for accuracy takes a longer time, especially if you're working with videos.

Parametric Tracking:- Since CNNs track and break down the entire image into smaller chunks, it can take a lot of system crunching to identify what part of the image, video or NLP framework led to the result and what portions don't contribute at all.

CNNs are nevertheless great tools that produce amazing results if executed and well planned out. They are used in techniques like signal



Working of a typical CNN

processing and image classification. They have also taken over computer vision techniques that have allowed for great advancements in aerial mapping drones, video analysis, turing operations, weather predictions and GPS.

They work so well because they split the content and make use of kernel like filters to detect every aspect of the input. In this case, a filter is just a matrix of weights that have been trained to identify outliers in the input and change accordingly as the simulations are run. These filters move each